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Social responsibility of major infrastructure projects in China

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Abstract

China has implemented a large number of major infrastructure projects (MIPs) over the last three decades. Social responsibility management is crucial for MIPs' sustainable development. What is social responsibility of major infrastructure (MIP-SR)? To answer this question, this article proposes the concept and key issues of major infrastructure projects' social responsibility (MIP-SR) and develops a comprehensive conceptual framework for MIP-SR, which covers three dimensions: (i) project life-cycle dynamics; (ii) stakeholder's heterogeneity and (iii) social responsibility interactivity. The three-dimensional framework provides a systematic framework for MIP-SR's academic research and practical implementation, which in turn promotes the sustainable development of MIPs.
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Keywords: Major infrastructure projects; Social responsibility; Project life-cycle; Stakeholders

1. Introduction

Major infrastructure projects (MIPs) are large-scale engineering facilities which provide fundamental public services for social production, economic development, and people's life, such as large-scale hydropower projects, high speed railways, expressway networks, gas pipeline projects and long-span bridges (Flyvbjerg, 2011). As the MIPs possess very important strategic positions in the national economy and social development, their social responsibility and sustainability have attracted widespread attention (Demetriades and Mamuneas, 2000; Flyvbjerg, 2014).

Large-volume MIPs have been designed, built and operated in China. Since the reform and open policy in 1978, a range of great projects such as the Three Gorges Dam, Qinghai–Tibet Railway and West–East National Gas Transmission Projects, have been built and operated. China has accumulated extensive experiences and gained impressive achievements on MIP construction and management (Shen et al., 2011a). However, it cannot be ignored that a series of public incidents about MIPs have aroused people's wide concerns in recent years (e.g. information disclosure problems in a serious accident[1]; violent clashes between the China Railway Group and local communities). The unexpected group incidents caused by environmental destruction or immigration settlement during construction are still common. These social responsibility deficiencies in MIPs have gone beyond the engineering projects themselves and widely triggered severe social concerns. Social responsibility has become one of the critical strategic factors related to the sustainable development of MIPs.

In the global context of sustainable development, China's MIPs are currently standing at a special and crucial period with strategic opportunities, environmental sensitivity and value reconstruction (Shen et al., 2011b). First, China's economy is already huge and is growing at the fastest rate compared to that

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[1] A bullet train accident happened in the rail line from Ningbo to Wenzhou in July 23, 2011, in which about 40 people were killed.

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of any developed nation in the last several decades, which gives great strategic opportunities for MIP implementation. Both national and local governments have proposed to focus on MIPs to stimulate domestic demand and maintain steady economic growth (Chen et al, 2013). Second, China's environmental problems are among the most severe of any major countries around the world and are mostly getting worse (Liu and Diamond, 2005). However, China is only ranked 118 out of 178 countries in the 2014 Environmental Performance Index (EPI).[2] Infrastructures are great projects which transform the nature of human activities and bring dramatic and permanent impacts to the environment. Since the public awareness of environmental protection, MIPs' great challenges on environmental protection and ecological balance have attracted wide concerns. Third, sustainable development becomes an important topic for governments, researchers and enterprises (McMichael, 2011; Sachs and Reid, 2006). Under the enormous pressures of natural resources and environment, MIPs' implementation development has presented a trend of valuable reconstruction from traditional GDP to sustainability. Above all, MIP development faces special and complex challenges in China nowadays and in such context, the pressing questions of its social responsibility need to be answered urgently.

This article analyzes the connotations of major infrastructure projects' social responsibility (MIP-SR). The main contribution of this article is to develop a systematic framework for MIP-SR, which covers three dimensions: (i) project life-cycle dynamics; (ii) stakeholder's heterogeneity and (iii) social responsibility interactivity. It is also hoped that the study can shed some light on the sustainable development of MIPs in emerging economies.

The remainder of this article is organized as follows. In the second section, the literature of infrastructure projects and social responsibility is reviewed. Next, the definition and key issues of MIP-SR are described. Dimensions and framework of MIP-SR are proposed in the fourth section. The final section presents commentary and conclusion, as well as limitations and some implications for future studies.

2. Literature review

2.1. Infrastructure management

Since a group of research centers on infrastructure management were established around the world,[3] infrastructure project management has received recent popular and academic attention. Many themes are involved, such as decision making, risk management, multi-project synergy and life-cycle cost (Flyvbjerg et al., 2009; Miller and Lessard, 2000; Priemus et al., 2008). It is noteworthy that complexity and sustainability are proposed as important issues on infrastructure management (Bosch-Rekveldt et al., 2011; Davies and Hobday, 2005; Flyvbjerg, 2014; Levitt, 2007).

[2] See http://epi.yale.edu/epi/country-rankings/.

[3] For example, Centre for Infrastructure & Construction Industry Development at Hong Kong in 2002, Centre for Major Program Management at Oxford in 2007, International Centre for Complex Project Management in 2008, Centre for Infrastructure Development at Manchester in 2010, etc.

2.1.1. Complexity of infrastructure

Major infrastructures can potentially bring huge financial investments, a long implementing period and multitudinous stakeholders. Infrastructures are thus complex engineering systems, which possess complicated uncertainties, potentially high risks and profound and lasting impacts on the economy, environment and society (Flyvbjerg et al., 2003; Miller and Hobbs, 2005). Complex engineering management needs to be understood and treated from the complex system perspective, in which it is important to capture the complex issues beyond the engineering itself (Ottino, 2004).

Baccarini (1996) proposes the concept of project complexity with differentiation and interdependency, which is developed by other scholars, such as Williams (2002), Remington and Pollack (2007) and Vidal and Marle (2008). Baccarini (1996) classifies project complexity into organizational complexity and technical complexity. Maylor (2003) then adds resource complexity as the third pattern; furthermore, other studies improve and enrich the patterns of complexity (Geraldi, 2008; Puddicombe, 2012; Xia and Lee, 2004). Otherwise, complexity analysis of major infrastructure is also developed (Alberts et al., 2004; Solis et al., 2013; Winter et al., 2006). Other topics of complexity of engineering are also explored, such as the organization interactivity (Antoniadis et al., 2011), large and complex networks (Pauget and Wald, 2013), and project communication and complexity (Senescu et al., 2012). In general, infrastructure complexity consists of technical complexity, organizational complexity and environmental complexity (Bosch-Rekveldt et al., 2011).

Peculiarly, the organizational and environmental components reflect the necessity and urgency of sustainability management on major infrastructure, which is associated with the complex dimensions of social responsibility. On one side, organizations in infrastructure have complex constitution, relationship and behaviors, which call for a higher strategic perspective (Thiry and Deguire, 2007). On the other side, the mainly complex environment involved in infrastructure management is about the social background and networking.

2.1.2. Sustainability of infrastructure

The complexity of major infrastructure brings the challenges of sustainability and social responsibility. Especially in the context of business globalization, technology integration and the whole life-cycle process, modern infrastructure project management must consider the economic, social and environmental impacts during the full life-cycle coverage (Levitt, 2007; Saynisch, 2008).

Numerous previous studies focused on the economic sustainability of infrastructure, including both projects' intrinsic objectives and their economic impacts on social economy (Atkinson, 1999; Demetriades and Mamuneas, 2000; Morrison and Schwartz, 1996; Munnell, 1992). As the success criteria of general projects, major infrastructures have to adopt the multi-objective strategy including cost, time and quality (Atkinson, 1999). However, major infrastructures have undertaken various functions which affect the nation and its society, thus it is more difficult to coordinate and deal with the conflicts from various objectives rather than conflicts from general projects (Flyvbjerg,

2011). Major infrastructures have huge financial investments, which require a long implementation period, and multitudinous stakeholders, whose scale effect and industrial linkage are very different from general projects. During a project's average life-cycle, consumption of natural resources and human resource input–output should also be considered for sustainable development (Parkins, 1978). Previous studies have shown that infrastructure projects have direct or indirect positive impacts on economic development (Esfahani and Ramirez, 2003; Munnell, 1992). With the theory of production function, major infrastructures could promote the national or local economy with scale effects (Hulten et al., 2006; Morrison and Schwartz, 1996). Also, infrastructure's impact on regional economic development in rural China has received attention among researchers (Fan and Zhang, 2004).

However, the systematic studies on the social and environmental impacts of major infrastructures are relatively less. Fortunately, many social and environmental issues on specific infrastructure projects have been proposed, such as the Three Gorges Dam's environmental and ecological impacts (Stone, 2008; Wu et al., 2003), and the social and environmental changes on Tibet Plateau caused by the Qinghai–Tibet Railway (Peng et al., 2007; Qiu, 2007).

2.2. Social responsibility and infrastructure

As an important part of sustainable development, social responsibility, especially corporate social responsibility (CSR) was proposed in the 1950s (Bowen, 1953; Frederick, 1994) and has since evolved in the last half of the century (Aguilera et al., 2007; Carroll, 1991, 1999). CSR has been an academic hotspot since the 1990s (Aguinis and Glavas, 2012; Campbell, 2007).

The academic studies on CSR are relatively systematic and plentiful. The essential subjects of CSR are social issues, environmental issues, ethical issues and stakeholder issues (Dahlsrud, 2008; Lockett et al., 2006). Scholars have focused on institutional, organizational and individual levels to analyze the reactive and proactive predictors of CSR actions and policies and their outcomes (Aguinis and Glavas, 2012). The implementation of CSR is constrained by internal factors including financial investments, organizational scales, economic performance and resources (Li and Zhang, 2010; McWilliams and Siegel, 2001). Market circumstances, governmental actions, industry developments, national characteristics and other external factors also influence the CSR performance (Aguilera et al., 2007; Matten and Moon, 2008). Additionally, indicators and assessment on CSR especially in the construction industry have also been considered (O'Connor and Spangenberg, 2008; Tam et al, 2006; Zhao et al., 2012).

With respect to infrastructure projects' social responsibility, scholars have carried out exploratory studies, especially in engineering construction. These studies mostly focus on the contractors' social responsibility in the construction phase of the engineering projects' life-cycle. For example, the contractor's environmental performance and strategic management are influenced by corporate policy, market circumstance and sub-contractor relationship (Zeng et al., 2003). Likewise, other project stakeholders are closely related to the engineering's environmental performance, such as governmental regulations (Qi et al., 2010; Tam et al., 2004) and public social pressure (Gluch, 2009). In spite of environmental protection, health care and safety insurance in construction projects are also involved (Fang et al., 2004; Oliver, 1997; Tam et al., 2004).

The social responsibility implementation in China attracts wide attention around the world (Moon and Shen, 2010). Since a number of major infrastructures are built and operated, the social responsibility and sustainability of MIPs have also received great interests. The environmental (Stone, 2008) and ecological (Wu et al., 2003) influences of the Three Gorges Dam Projects were reported in *Science*. The Qinghai–Tibet Railway's social responsibility issues were also published in *Nature* (Qiu, 2007), including green construction, environmental protection, ecological balance, immigration and social development. Peng et al. (2007) describes the environmentally-friendly behaviors during the construction life-cycle of the Qinghai–Tibet Railway and states the infrastructure's effect on the local transportation and tourism industry.

In summary, the complexity of an infrastructure, especially organizational complexity and environmental complexity, brings challenges for the sustainability and social responsibility of MIPs (Miller and Hobbs, 2005). Previous studies about sustainability are relatively scattered, most of which just focus on one aspect of social responsibility, like economic impacts, and environmental risks or only analyze some specific infrastructure projects and their conclusions are not universally applicable. That is, existing studies on the social responsibility of major infrastructures are fragmented and still not systematic. Fortunately, previous corporate social responsibility (CSR) literature provides necessary theoretical bases; meanwhile, a series of building or operating infrastructure project cases have been successfully achieved in China, which provides strong practical experience. Briefly, it is necessary and meaningful to integrate major infrastructure management and social responsibility studies to satisfy the requirement of MIP sustainable development. Thus, the primary and principal task is to analyze the conception of MIP-SR and establish a comprehensive and systematic theoretical framework for MIP-SR.

3. Social responsibility of major infrastructure projects

3.1. Definition of MIP-SR

The essential characteristic of social responsibility is the willingness of an organization to incorporate social and environmental considerations in its decision making and be accountable for the impacts of its decisions and activities on the society and environment (ISO, 2006). The organizations implementing the social responsibility in MIPs are direct participants (governments, contractors, suppliers of equipment and/or materials, designers, employees) and other stakeholders (public community, non-government organizations). The social responsibility activities need to be managed through projects (Salazar et al., 2012), especially practiced through their life-cycle. As MIPs hold a pivotal position in the state and local economy, both transparent and ethical behaviors in MIPs

contribute to sustainable development. Because sustainable development is about the economic, social and environmental goals common to all people, it can be used as a way of summing up the broader expectations of society that needs to be taken into account by organizations seeking to act responsibly (ISO, 2006). Hence, an overarching objective of MIPs' social responsibility should contribute to sustainable development. In addition, the project behaviors should be in compliance with applicable laws of states and local governments which should be consistent with international standards and industrial norms. It is also implied that the social responsibility of major infrastructures is integrated among the participants, who have exercised the idea and taken into account the interests of stakeholders.

To sum up, MIP-SR could be proposed as:

Social responsibility of major infrastructure projects involves the policies and practices of the stakeholders participated through the whole project life-cycle that reflects responsibilities for the well-being of the wider society.

3.2. Conflicts in MIP-SR

Despite the conflicts of time limitation, budget constraint, quality requirement and basic functions (e.g. produce electricity, open to traffic and operation) like general projects, MIPs are often additionally expected to promote local economic development significantly, improve people's life substantially, allocate public resources rationally and adapt national strategic deployment smartly. Moreover, besides the major economic benefits, there are conflicts in MIP-SR. For instance, reformation and destruction to the environment and impacts to the future generations can by no means be ignored. And it is extremely crucial to effectively reconcile and integrate the conflicting needs in MIP-SR. These conflicts lead to the interactivity of MIP-SR and can be reflected in resource consumption, environmental impact and project performance, and stakeholders' appeals.

3.2.1. Conflicts of resource consumption, environmental impact and project performance

Most MIPs in China have been conducted by the Chinese government, which require huge capital investments and enormous resource consumption during the construction and operation. After completion, the projects would lead to a new allocation of resources in the local economy. Not only the current consumption in the construction and operational phases is concerned, but also the next centurial or even millennial impact of the major infrastructures should be considered. No matter whether the impact is positive or negative, it is far-reaching. For instance, the Dujiangyan Irrigation System which was built in 256 B.D. has overcome the drought and flood disasters for over two thousand years and still has helped grow Chengdu Plain's prosperity (Li and Xu, 2006). Inversely, after the disaster of Chernobyl nuclear power station in Ukraine (1986), the radiated pollution covered an area of about 150,000 km^2 and severely affects at least 6.9 million people's health and safety.

Furthermore, as MIPs require more investment and consume more public resources than general projects, their impact to local economic development, social progress and environmental changes is profound and usually irreversible. Although it is consistent with the pursuit of project performance to focus on the current efficiency of public resources, there might be conflicts in the post-operational resource allocation, impact to the future environment and current project performance (such as time, cost and quality). Consequently, a multi-dimensional, higher-level and longer-term requirement of major infrastructure should be carried into effect through the whole project life-cycle, which calls for strict quality and standards with additional time and cost.

3.2.2. Conflicts of stakeholders' appeal

Unlike general projects, MIP stakeholders' conflicts mainly concentrate on the governments, construction corporate and the public. Decision-making, planning, management and coordination are processes of multi-stakeholders' interaction including strategic synergy, business collaboration, procedure coordination and information consultation. There are complex conflicts and interactive stakeholders' relationship inside and outside the infrastructure. The multi-stakeholders' conflicts sometimes intensify and escalate in a project life-cycle, which might stir up unwanted social public incidents.

In China, most MIPs are often initiated by the central government (e.g. Ministry of Communications, Ministry of Water Resources, National Development and Reform Commission) or local governments; and specialized state-owned companies (e.g. China Three Gorges Corporation, China Railway Construction Corporate) are organized to design, construct and operate; what is more, the governments' supervisory departments (such as the Ministry of Environmental Protection) are also involved. Since every participant has their own interest and agenda, contradiction usually arises, such as the conflict of investment and resource allocation in a multi-level government, disputation of responsibility and authority, supervision and coordination. For instance, the Three Gorges Corporation was listed on the Ministry of Environmental Protection's violating enterprises in 2005. Then a debate lasted for three months and was eventually settled with the mediation of the National Development and Reform Commission. What's more, in the context of strong government and weak regulations, some organizations' alienation behaviors have seriously affected the social responsibility performance of MIPs, such as excessive administrative intervention, public power alienation, opportunistic decision making, bribery and corruption.

The public stakeholders of MIPs are usually with a wider range and more complicated levels (Li et al., 2013). According to the impacts from the projects, the public stakeholders could be classified into two categories: local community and general public. The local community is directly affected by the MIPs, and confronts immigration, demolition, etc. If environmental pollution or safety accidents occur, people in the local communities would be the direct victims. On the other hand, they may also gain an improvement in their income and living condition from the local economic prosperity of MIPs. The general public is not

directly affected by the MIPs, but care for the economic, social and environmental performance of MIPs. Thus, the general public's evaluation and views would unbiasedly influence the government image and credibility which in turn may intervene with MIP management (Li et al., 2013; Ng et al., 2013).

3.3. Key issues of MIP-SR

The general issues of social responsibility involve human rights, labor practices, environmental protection, fair operating practices, consumer issues, community involvements and developments (ISO, 2006). In China, CSR is mainly extracted as economic responsibility, legal responsibility, environmental protection, customer orientation, people focused, charity, employment, good faith, social stability and progress (Xu and Yang, 2010). However, compared with CSR, MIP-SR has its own unique issues. Especially in China, the fast-developing country with great pressures of economic development, social progress and environmental protection, MIPs of China are now at a critical period of strategic opportunity, environmental sensitivity and valued reconstruction.

3.3.1. Immigrant settlement

MIP immigration is a social problem, which comes down to social equity, regional disparity, social stability and ecological balance. For example, there have been about 160 million migrants due to water infrastructure construction in China in the last decades. The conflicts among government behaviors, public requirements and economic developments are common, such as migrant adaptability in residential areas and resettlement compensation. Hence, appropriate compensation and settlement policies are critical to gain public trust and to support the initial phase of a project's life-cycle (Fincher, 1991).

3.3.2. Pollution control

The MIPs consume a great amount of natural resources, which might lead to construction trash, air pollution, water pollution and land pollution. Construction activities have significant adverse impacts on the environment which typically include dust and gas emissions (Zhao et al., 2012). In China, the construction trash accounts for about 40% of the total solid waste generation, reaching about 300 million tons per year (Wang et al., 2010); China's energy infrastructure is expanding at an unprecedented speed, most of which use coal for power generation and over-produce green gas emissions (Liu et al., 2008).

3.3.3. Ecological protection

As MIPs would significantly change the local natural environment (land, ocean, river, air), the ecological balance can easily be broken. Habitat fragmentation due to construction, including both the reduction and alteration of its spatial configuration, has been generally recognized as the primary cause for the loss of biodiversity and ecosystem services (Wu et al., 2003). Thus the natural vegetation might be devastated and the habitat of wild animals is damaged in local regions. Water infrastructures usually change ecological processes such as nutrient and water recycling. The project may inundate vast expanses of land upstream, which may overwhelm vegetation and significantly reduce the main water supply that could seriously influence aquatic plants and fish species downstream. The Three Gorges Dam Project's ecological fragmentation has riveted the world's attention (Shen and Xie, 2004). It is reported that approximately 560 species of rare terraneous plants are submerged when the reservoirs of the Three Gorges Dam are filled up. Moreover, other major infrastructures such as railway projects could severely impact the local wild animals' migration routes (Peng et al., 2007). According to reports by the state-run Chinese media, as many as 1500 antelopes' migration and breeding activities are severely affected by the suspended construction of the Qinghai–Tibet railway on the plateau (Qiu, 2007).

3.3.4. Occupational health and safety

Since most MIPs are built at desolate places, the harsh outdoor work environment causes occupational diseases more easily than other industries. Complex participation of multi-subjects is involved at the construction scene of MIPs, where particular environmental sensitivity and complex technology should not be overlooked. Thus, the occupational health and safety risk of MIPs are quite high and accidents on site often occur. There are about 60 thousand project-related accidents which happened on site, and at least 60,000 fatal accidents occur each year at construction sites around the world, which is equivalent to one death every 10 min (Martínez Aires et al., 2010). It is also reported that about 35% of all construction workers affirm that their work entails a health risk according to the fourth European survey of work conditions (OSHA, 2007). Compared with other industries, the mortality and injury rates of construction project practitioners are about 50% higher (Loosemore and Phua, 2011). Through the life-cycle of MIPs, these health impacts and safety events would not only cause catastrophic consequences to employees, but also influence the social responsibility (such as the governmental image and contractor's reputation) and thus increase the project cost.

3.3.5. Anti-corruption

There are many scandals about corruption in MIPs, especially in the bidding process during the initialization phase. In the Bribe Payer Index of Transparency International, public works and construction receive the worst score.[4] This is due to the particular characteristics of MIPs, which are usually large, unique, complex and full of conflicts; therefore they are difficult to benchmark for cost and time. These make it easier to hide and inflate additional expenditure. MIPs also form a fragmented industry, often involving multi-stakeholders, complicating the tracing of payments and the diffusion of standards and practices. In such context, corruption threatens the quality and safety standards of infrastructures used by the public every day, which, as witnessed by the many deaths from earthquakes in highly corrupted countries, has a very real impact on human lives. Generally, corruption in construction has penetrated into every

[4] See http://bpi.transparency.org/.

phase of projects and brought serious negative effects on sustainable development (Loosemore and Phua, 2011; Sohail and Cavill, 2008); it especially tends to influence governmental decision-making (Kenny, 2009), which not only causes project quality problems and security incidents, but also results in public complaints and great damage to government image and credibility.

3.3.6. Disaster prevention and mitigation

According to statistics, over 95% of natural disasters occurred in developing countries (Campher, 2005), where most disasters are closely related to the lack or imperfection of infrastructure. Developing major infrastructures plays a pivotal role in effectively preventing and mitigating disasters (Weichselgartner, 2001). MIPs could improve resilience in areas at substantial risks from catastrophic events (Guikema, 2009). For instance, dams could help rein in flooding; well-developed expressway networks could deliver effective routes in emergency.

3.3.7. Poverty eradication

MIPs are supposed to generally improve social productivity (Morrison and Schwartz, 1992), so as to increase the local employment and eradicate poverty. When there is growth in MIPs, especially transportation infrastructures such as high-speed railways and expressway networks, productivity growth changes disproportionately in relevant industries such as the vehicle-intensive sector (Fernald, 1999). Intuitively, MIPs facilitate the productivity of social economics so that relevant industrial employment would improve and the local economic would also be stimulated as well whereas, the social resource reallocation brought by MIPs might also lead to secondary poverty in particular situations (Cernea, 1988). It is also realized that transportation investment does not always appear unusually productive, sometimes even slowing down productivity (Fernald, 1999).

4. A systematic framework of MIP-SR

The social responsibility of major infrastructure projects (MIP-SR) covers complex conflicts and a series of key issues as stated above. However, the project concept and the context will evolve over time through successive episodes (Miller and Hobbs, 2005), and multiple stakeholders and varieties of issues are involved inside and outside of MIPs. Thus, how does the MIP-SR evolve through its life-cycle? Who will bear the responsibility? What spectra of social responsibility should be undertaken? Generally, from the project life-cycle dimension, MIP-SR should cover a long project life-cycle which concludes distinctive phases; from the stakeholders' dimension, MIP-SR involves more heterogeneous participants and many extensive stakeholders, and faces great intricate conflicts of interest; and from the social responsibility dimension, more than general projects, as such, MIPs need to fulfill a high level of responsibility.

4.1. Research methodology

This article started with a review on project management, infrastructure management, social responsibility and business ethics literature. Based on the project management body of knowledge, stakeholder theory and corporate social responsibility studies, the conceptual model was proposed (Zeng, 2012), and further refined through a series of interviews with scholars and engineers who are engaged in infrastructure studies and implementation in China.[5]

4.2. Project life-cycle dimension

In fact, the implementation of social responsibility should run through the whole project life-cycle, which contains an initialization phase (conception, feasibility, identification), plan and design phases, a construction phase and an operational phase (Miller and Hobbs, 2005; Munns and Bjeirmi, 1996). The activities of organizations in these phases have direct economic, social and environmental impacts at the project level, so as to influence sustainability (Keeble et al., 2003). As MIPs are advancing along with the project life-cycle, both primary participants of projects and key issues of social responsibility are changing dynamically. Fig. 1 shows the stakeholders and issues of MIP-SR throughout its life-cycle.

On one side, the participants or extensively the stakeholders in MIPs play different roles in each phase of a project's life-cycle. As shown in Fig. 1, the key roles in the initialization phase are governments, the general public and the local communities. Because MIPs are almost initialized by governments and built for the general public, these stakeholders should discuss the necessity of MIPs during the conception and participate in the analysis of the projects' feasibility. Thus, proper information disclosure and participation mechanisms in this phase are of great importance. And the governments' organizational alienation behaviors (corruption) in the decision making of MIPs need to be tackled. Furthermore, designers are supposed to adapt social responsibility; contractors and suppliers of equipment or materials are in charge of construction works in the construction phase.

On the other side, social responsibility takes on different issues in different life-cycle phases. For example, public participation, decision-making transparency, and information disclosure can attract more attention during the initialization phase. In the design and plan phases, MIP-SR mainly involves green innovation pursuit, process validity and environmentally-friendly design. Additionally, the employees' health care, the site safety management, urgent accident response, environment protection and resource conservation are focused on during the construction phase. Public event management and ecological balance are distinctive in the operational phase.

[5] These interviews are semi structured and conducted in the first half of 2013. Scholars are from Harbin Institute of Technology, Huazhong University of Science and Technology, Nanjing University, and Tongji University, etc. Engineers are working (or used to work) at the China Railway Engineering Corporation, Hong Kong–Zhuhai–Macau Bridge Project, etc.

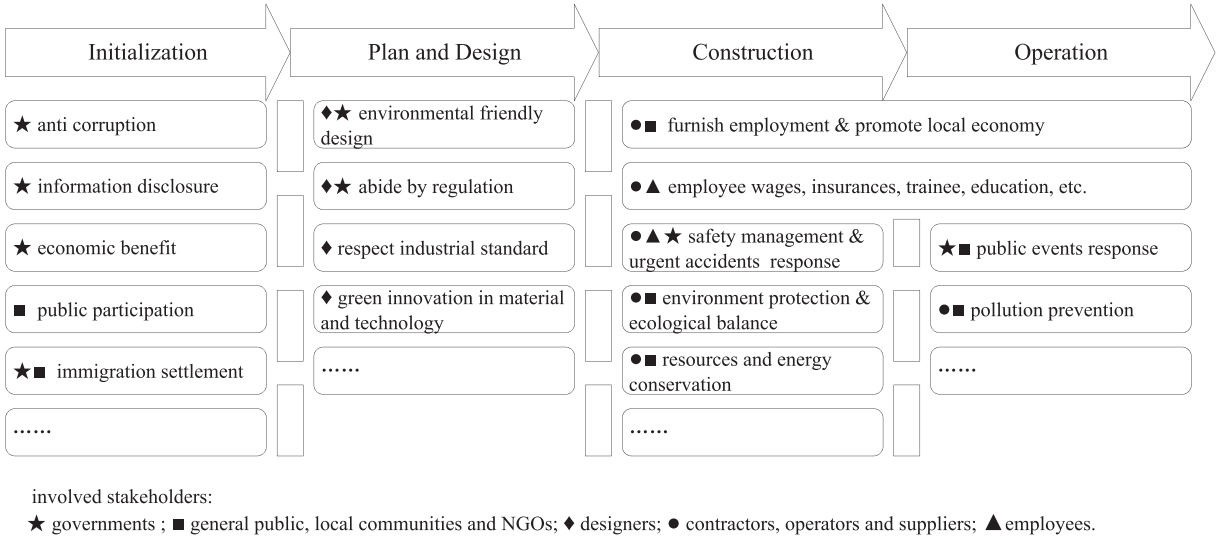


Fig. 1. Project life-cycle of MIP-SR.
It is noteworthy that this figure is tentative and open, and the issues in the different phases are not collectively exhaustive, as the practices and studies of both infrastructure management and social responsibility are diversified and dynamically developing when more issues would absolutely appear in the future.

4.3. Stakeholder dimension

Project stakeholders are individuals and organizations that are actively involved in the project, or those whose interests may be affected by the execution or completion of a project. The MIPs have a wide variety of salient stakeholders, including governments, contractors, suppliers, designers, employees, the general public, local communities and non-governmental organizations (NGOs). They can also exert distinctive influence over the project's objectives and outcomes. Therefore, the key of stakeholder management in MIP-SR is to balance the interest of every party (Freeman, 1984).

As the impacts from different stakeholders are distinguishable, the stakeholders' effects on the implementation of MIP-SR are heterogeneous. There is synergy in the implementation of social responsibility among organizations; in other words, the inside and outside stakeholders interact, coordinate, promote and combine to perform the social responsiveness.

It is necessary to classify the stakeholders in MIP-SR. There are some scholars who distinguish the stakeholders from varieties of perspectives in CSR (Barnett, 2007; Buysse and Verbeke, 2003; Clarkson, 1995; Harrison and Freeman, 1999). Based on the contractual relationship with the corporate, Charkham (1992) divided organizations into contractual stakeholders (shareholders, employees, clients, distributors, suppliers, creditors) and public stakeholders (consumers, supervisors, governments, media, local communities). In the field of environmental management, Darnall et al. (2010) divided stakeholders into direct stakeholders (consumers, purchasers, suppliers, managers, employees) and indirect stakeholders (environmental groups, communities, labor union, environmental protection bureaus). Huang and Kung (2010) classified the stakeholders as: external interest groups (governments, creditors, consumers, competitors), internal interest groups (shareholders, employees) and mediate interest groups (environmental organizations, accounting firms). Learning these scholars' studies and considering the MIPs' characteristics, the stakeholders of MIP-SR could be classified into two categories: direct–internal–contractual groups including governments (as most infrastructures are dominated by the states or local governments), designers, contributors, suppliers of equipment and materials, operators, employees, and indirect–external–public groups including the general public, local communities and NGOs as shown in Fig. 2.

4.4. Social responsibility dimension

MIPs are expected to undertake great social responsibility by both the government and public in the conception phase and have great impacts on the national economy, people's life and local environment in the construction and operation phases. Especially, MIPs always tend to last a very long period and require substantial investment (finance, labor, and natural resources); the economic, social and environmental changes of the local area are perennial, immense and irreversible. As a result, MIPs cover multiple spectra of social responsibility. Moreover, MIP-SR's formation is much complex than what different stakeholders have in their own perspectives, such as profitability, morality, philanthropy, political and take varieties of social responsiveness during the life cycle of MIPs. Above all, it is necessary to classify the different spectra of social responsibility so as to analyze the mechanism of MIP-SR.

Scholars have different views on the spectra of social responsibility. Carroll (1991) proposed a classical four-component pyramid model of social responsibility: economic, legal, ethical and philanthropic components. Shaw and Post (1993) modified the pyramid model and pointed out that philanthropy should be one part of ethical responsibility. As the government plays an important role in the implementation of social responsibility,

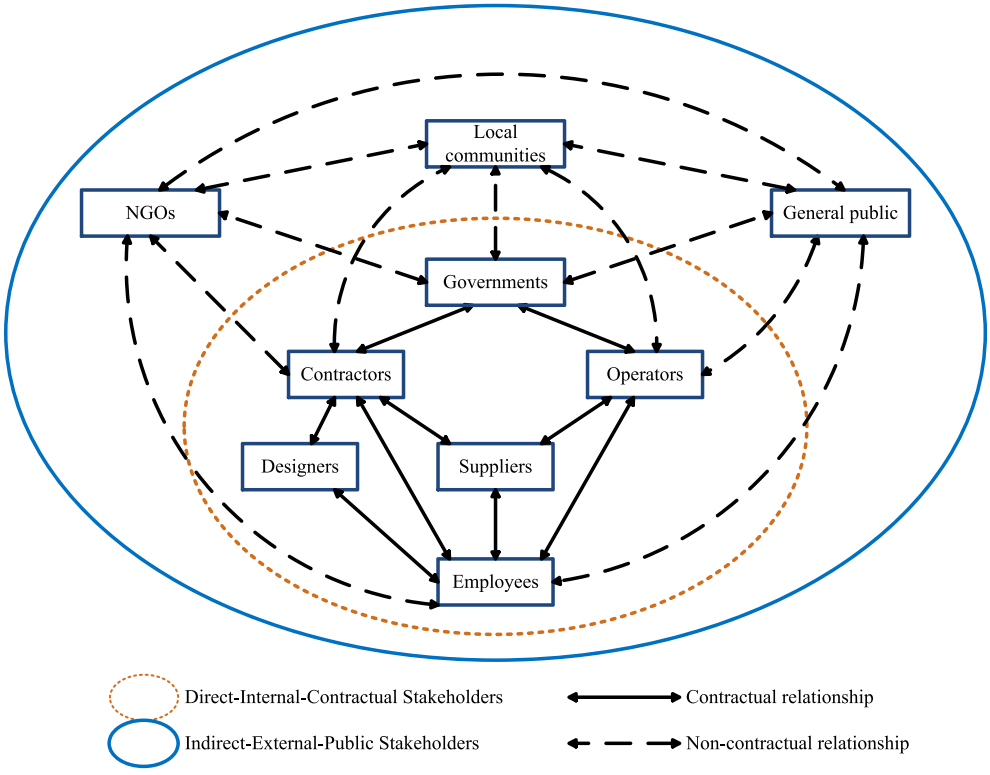


Fig. 2. Stakeholders of MIP-SR.
The contractual or non-contractual relationship in Fig. 2 does not just imply that there is a specific contract agreement between the stakeholders or not, but indicates the relationship of principle–agent constraints. What's more, these stakeholders in Fig. 2 are relatively stable, especially in the context of China.

Scherer and Palazzo (2011) pointed out that it was necessary to integrate the organization's social responsibility and political role effectively. Garriga and Mele (2004) divided social responsibilities into economic profits, political performance, social demands and ethical values. Based on previous studies and the crucial status of MIPs in the national or local economic development, environmental protection, political stability and social progress, we propose that there are four spectra in MIP-SR: economic, legal, ethical and political responsibilities.

4.4.1. Economic responsibility

MIPs are also great public service systems to maintain or improve national or local economy and are supposed to bring economic benefits in three levels as below. At the macro level, the infrastructure has a direct positive effect on the national economic growth as a kind of production factor investment (Aschauer, 1989; Fan and Zhang, 2004); it also promotes the GDP indirectly as common goods (Straub, 2011). At the meso level, there are many industries related to the MIPs. And as a (quasi) public good, the infrastructure has a relatively weak-competitive and local exclusive market which has great scale and network effects (Behrens, 2011; Zheng and Kahn, 2013). At the micro level, it is obvious that the infrastructure investment could create job opportunities and reduce unemployment (Leigh and Neill, 2011), which also increase local personal incomes.

4.4.2. Legal responsibility

It is also important for MIPs' participants to perform in a manner consistent with the laws or various regulations promulgated by governments (Carroll, 1991). Besides the laws and regulations, MIPs should also be implemented within the framework of industrial norms and international standards. In practice, the corruption of infrastructure in developing countries (Bardhan and Mookherjee, 2000; Kenny, 2009), engineering safety risk problems caused by noncompliance of rules and other illegalities or irregularities have attracted much attention from governments, scholars and the public. Thus, fulfilling the legal obligations and meeting the regulatory requirements are very fundamental objectives of social responsibility.

4.4.3. Ethical responsibility

Ethical responsibility in MIPs embraces activities and practices that are expected or prohibited by society members, where three main issues (human rights, social philanthropy and environmental protection) are involved. First, human rights concern both inside and outside MIPs. Inside the projects social responsibility mainly covers employees' issues such as occupational health and safety (OHS), wages and welfare insurance, staff employment fairness, vocational education and training; while outside the project ethical issues include prevention of safety hazards to the community, harmonious communication and good information disclosure with the public. Second, social

philanthropy refers to charity activities to help local disadvantaged groups with financial support and to build community welfare facilities. Finally, environmental protection issues include green design and construction of infrastructure, reduction of energy cost, conservation of natural resources, and prevention and treatment of pollution.

4.4.4. Political responsibility

As discussed above, most MIPs are propelled by the government and could bring great impacts on the national economy and people's livelihood; hence, the political responsibility of MIPs is much greater than CSR. The MIPs' effect radiates around the local area much broader than CSR. MIPs could raise employment, improve the physical and mental health of a community (Ludwig et al., 2012) and increase the public's happiness, so as to promote social stability and progress, especially in China (Xu and Yang, 2010). Moreover, MIPs play an important role in the local poverty eradication or the immigration problems (Fincher, 1997; Murphy, 1990).

A caveat is that the four spectra of social responsibility are not absolutely independent but have overlaps among them (Schwartz and Carroll, 2003). For example, the environmental protection of MIPs is related to both legal responsibility (environmental violation) and ethical responsibility (harmful factors to local communities). Therefore, the interactivity of different spectra provides the coupling effect of MIP-SR, which is discussed in Section 4.4.

4.5. A three-dimensional model of MIP-SR

In summary, the social responsibility of MIPs has three dimensions and presents three hallmarks: project life-cycle dynamics, stakeholders' heterogeneity and social responsibility interactivity. Fig. 3 demonstrates the three-dimensional coordinates of MIP-SR, in which the x-axis refers to the project life-cycle, the y-axis refers to the stakeholders' heterogeneity and the z-axis refers to the social responsibility dimensions.

Fig. 3 also shows two examples of MIP-SR: government social responsibility in the initialization phase and contractors' social responsibility in the construction phase. Their spectra are different from each other and we could recognize that governments mainly undertake political responsibility in the initialization while contractors are responsible for implementing the economic, ethical, legal, and political spectra according to the sequence of priority during the construction phase. By this analogy, we could describe other stakeholders' social responsibility spectra at each phase of the MIP's life-cycle in the framework. In general, the MIP-SR evolves as the project cycle's advancement, distributes according to the stakeholders status and relationship, and interacts among economic, legal, ethical and political components.

Furthermore, in different life-cycle phases of MIPs, each stakeholder undertakes different spectra of social responsibility. And the implementation of social responsibility in complex MIPs involves interactive and synergetic actions among the direct–internal–contractual groups and indirect–external–public groups. These interactive and synergetic actions could be regarded as *Coupling*, which presents mutually dependent, coordinating, promotive and dynamic relationships. The effect of coupling in MIP-SR lies on the instrumental, relational and moral motives, which interplay within individuals, organizations and at national levels (Aguilera et al., 2007). Specifically, throughout the project life-cycle, stakeholders involved in the MIPs share the resources and construct networks, coordinate the overlaps of different social responsibility components, impel the integration of project systems and consequently achieve the synergetic effect of MIP-SR (Ansoff, 1987).

5. Discussion and conclusion

MIPs provide fundamental public services to social production, economic development and people's life. In China, MIPs have been a crucial support of national economy and social progress. However, a series of public events caused by a lack of social responsibility have gone beyond the infrastructure projects themselves and evolved into serious social problems. These problems could be extracted and summarized into several issues: immigration settlement, project pollution control, ecological protection, occupational health and safety, anti-corruption, disaster prevention, and mitigation, and poverty eradication. Furthermore, MIP-SR presents the characteristics of project life-cycle dynamics, stakeholders' heterogeneity and social responsibility interactivity. This paper has offered a systematic overview of previous research on social responsibility, proposed the concept of MIPs' social responsibility, and established the three dimensional model of MIP-SR. Our study adapts both a theoretical review and practical experiences (particularly in China) and explores a conceptual framework for the social responsibility of MIPs. The "life cycle–stakeholder–social responsibility" model illustrates the characteristics of long life span, multi-stakeholders and complex responsibilities. This framework is supposed to change the partial, scattered, non-system state of traditional research and provide new ways of thinking on the social responsibility of MIPs.

In the three-dimensional framework, the project life-cycle dimension globally covers conception, feasibility, identification, design, construction, delivery and operation. In the different phases, the focuses on stakeholders and issues of MIP-SR are significantly different and show a dynamic process with the promotion of projects. The stakeholder dimension has heterogeneity not only in various types of infrastructures but also in different phases of one project. Based on the relative studies on CSR, the stakeholders in MIP-SR could be classified into direct–internal–contractual (groups including governments, designers, contributors, suppliers of equipment and materials, operators and employees) and indirect–external–public groups (including general public, local communities and NGOs). Additionally, MIPs are the great public service systems, which undertake economic, legal, ethical and political responsibilities. The four spectra of social responsibilities in the third dimension are not absolutely independent, and the overlaps among them lead to the interactivity of MIP-SR.

Overall, the three-dimensional framework demonstrates the coupling mechanics of MIP-SR. The evolution through life-cycle, the synergy of various stakeholders, and the interactivity among spectra deconstruct the complexity of MIP-SR and bring

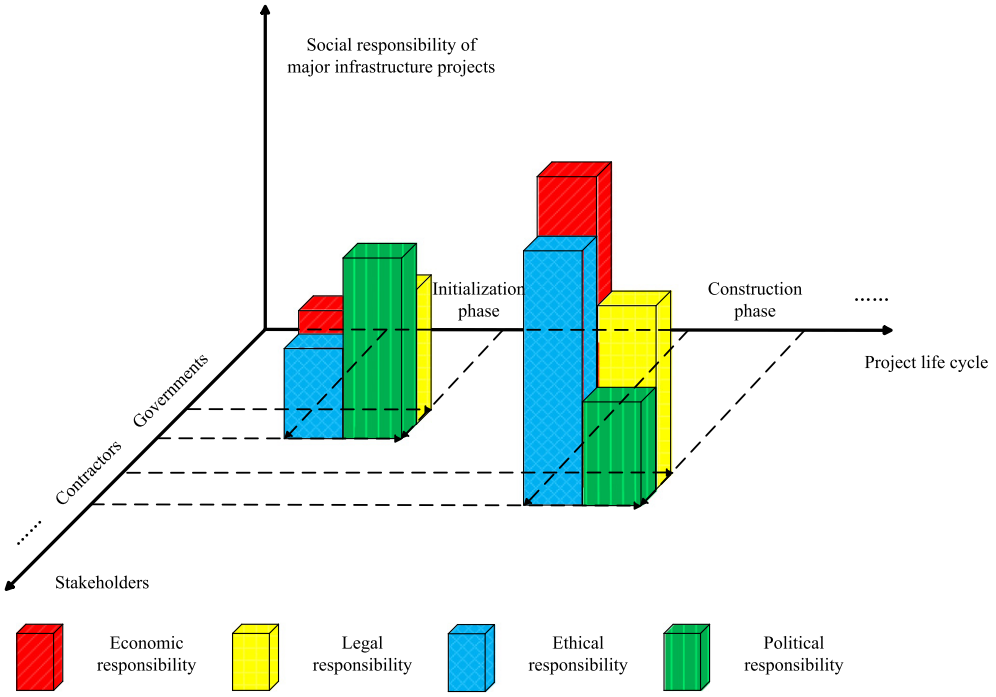


Fig. 3. Three dimension model of MIP-SR.

insights into the conflicts among the implementation of MIP-SR and sustainability. This conceptual framework hopefully provides systematic guidance for MIP-SR's academic research and practical implementation, which in turn will promote the sustainable development of MIPs. Specifically, this framework is supposed to provide a direct guideline to the evaluation or standards of social responsibility performance of MIPs. Thus, various stakeholders could be promoted or regulated to implement the policies and practices of social responsibility through the project episodes. What's more, although the mechanism of the three-dimensional framework is still needed for empirical analysis, it could still help the participants to insight the interactivity with other stakeholders in the context, so as to improve the decision making of social responsibility. For instance, governments should dynamically take appropriate behaviors (such as promoting rational information disclosure about the MIPs, choosing defensive, reactive or proactive responsiveness to public after emergent accidents) according to the interactivity of social responsibility in a specific phase of a project life-cycle.

Our study has several limitations but also reveals several promising areas for future research. First, this paper mainly focuses on major infrastructure projects in China, but whether the study could be extended to general engineering projects and to other countries around the world or not, is required to be validated in the future. As mentioned in Section 2.2, national circumstances, cultural characteristics and other factors would greatly influence the social responsibility fulfillment. Second, our study is based on literature review and theoretical analysis, in which the conclusion has been supported by a small number of cases in China, but the practical knowledge is still anecdotal for now. Clearly, empirical examinations are quite necessary to be adopted in the next step. Third, the three-dimensional model shows a basic framework of MIP-SR; nevertheless the detailed content of social responsibility is extended to consummate, especially the coupling effect of three dimensions in the model should be further explored both theoretically and empirically in the future. Thus the framework of MIP-SR could be used to improve the application of sustainable development more efficiently.

Conflict of interest

The authors have declared that no conflict of interest exists.

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